PART A — QUESTIONS

AF14-001
Question:
Solve: $16x + 7 = 103$
A) 5
B) 6
C) 7
D) 8

AF14-002
Question:
If $14n - 6 = 92$, what is n?
A) 5
B) 6
C) 7
D) 8

AF14-003
Question:
Simplify: $7(2x + 5) - 4x$
A) $10x + 35$
B) $14x + 35$
C) $10x + 5$
D) $18x + 35$

AF14-004
Question:
A delivery service charges $28 per stop plus a $15 scheduling fee. Which expression represents the total charge for s stops?
A) $28s + 15$
B) $15s + 28$
C) $43s$
D) $28 - 15s$

AF14-005
Question:
If $f(x) = 11x - 20$, what is $f(6)$?
A) 36
B) 46
C) 66
D) 86

AF14-006
Question:
Which ordered pair satisfies the equation $y = 9x - 2$?
A) (1, 8)
B) (2, 16)
C) (3, 25)
D) (4, 35)

AF14-007
Question:
Solve: 15(x - 3) = 90
A) 7
B) 8
C) 9
D) 10

AF14-008
Question:
A pattern begins: 12, 25, 38, 51, ... What is the next number?
A) 62
B) 63
C) 64
D) 65

AF14-009
Question:
Solve: x/24 = 3
A) 27
B) 48
C) 72
D) 96

AF14-010
Question:
A line has a slope of -11 and crosses the y-axis at 8. Which equation represents the line?
A) y = 8x - 11
B) y = -11x + 8
C) y = 11x + 8
D) y = -8x + 11

AF14-011
Question:
If 6 containers hold 138 pieces, how many pieces are in 4 containers at the same rate?
A) 84
B) 88
C) 92
D) 96

AF14-012
Question:
Solve: 100 - 15x = 40
A) 3
B) 4
C) 5
D) 6

AF14-013
Question:
Which expression is equivalent to 50y - 21y - 12?

A) 29y - 12
B) 71y - 12
C) 29y + 12
D) 50y - 33

AF14-014
Question:
A supply room starts with 420 forms. Workers use 14 forms each day. Which equation gives the number of forms F left after d days?
A) F = 420d - 14
B) F = 14d + 420
C) F = 420 - 14d
D) F = 14d - 420

AF14-015
Question:
If y = -13x + 120, what is y when x = 8?
A) 16
B) 24
C) 104
D) 224

AF14-016
Question:
Solve: 22x - 25 = 13x + 56
A) 7
B) 8
C) 9
D) 10

AF14-017
Question:
Which value of x makes 12x + 3 greater than 99?
A) 7
B) 8
C) 9
D) 6

AF14-018
Question:
A table shows a linear relationship.

x: 0, 1, 2, 3
y: 5, 19, 33, 47

What is the rule?
A) y = 5x + 14
B) y = 14x + 5
C) y = 19x - 14
D) y = x + 5

AF14-019
Question:
Solve: 11(x + 5) = 8x + 82
A) 7
B) 8
C) 9
D) 10

AF14-020
Question:
If a = -9 and b = 7, what is 3a + 4b?
A) -55
B) -1
C) 1
D) 55

AF14-021
Question:
Which point has a positive x-coordinate and a negative y-coordinate?
A) (-10, 4)
B) (10, -4)
C) (-10, -4)
D) (10, 4)

AF14-022
Question:
A worker completes 144 checks in 12 minutes. At the same rate, how many checks can the worker complete in 9 minutes?
A) 96
B) 104
C) 108
D) 120

AF14-023
Question:
Solve for r: M = 18r + 36
A) r = M - 36
B) r = (M - 36)/18
C) r = 18(M - 36)
D) r = M/18 + 36

AF14-024
Question:
What is the slope of a line that goes through (2, 5) and (8, 53)?
A) 6
B) 7
C) 8
D) 48

AF14-025
Question:

Simplify: $36x + 8 - 25x - 17$
A) $11x - 9$
B) $61x - 9$
C) $11x + 25$
D) $61x + 25$

AF14-026
Question:
If $g(x) = x^2 - 8x$, what is $g(12)$?
A) 36
B) 48
C) 96
D) 144

AF14-027
Question:
A sequence follows the rule “subtract 6, then multiply by 4.” If the first term is 14, what is the second term?
A) 8
B) 24
C) 32
D) 56

AF14-028
Question:
Solve: $14(x - 5) + 8 = 92$
A) 9
B) 10
C) 11
D) 12

AF14-029
Question:
A line crosses the y-axis at -13 and rises 6 units for every 1 unit to the right. Which equation represents the line?
A) $y = -13x + 6$
B) $y = 6x - 13$
C) $y = -6x - 13$
D) $y = 13x + 6$

AF14-030
Question:
If $16/24 = x/72$, what is x?
A) 36
B) 42
C) 48
D) 54

AF14-031
Question:
Solve: $-16x + 32 = -96$

A) -8
B) -4
C) 4
D) 8

AF14-032
Question:
A stockroom starts with 1,800 fasteners. Each job uses 90 fasteners. Which expression gives the number of fasteners left after j jobs?
A) 1,800 + 90j
B) 1,800 - 90j
C) 90j - 1,800
D) 1,800 ÷ 90j

AF14-033
Question:
Which equation has x = 19 as its solution?
A) 2x + 10 = 46
B) 4x - 15 = 61
C) x/19 + 6 = 8
D) 3x - 7 = 48

PART B — ANSWER KEY + EXPLANATIONS

AF14-001 — Correct: B — Explanation:
Solve 16x + 7 = 103. Subtract 7 from both sides: 16x = 96. Divide by 16: x = 6. The most common mistake is subtracting 7 correctly but forgetting to divide by 16.

AF14-002 — Correct: C — Explanation:
Start with 14n - 6 = 92. Add 6 to both sides: 14n = 98. Divide by 14: n = 7. A common mistake is subtracting 6 instead of adding it.

AF14-003 — Correct: A — Explanation:
Distribute first: 7(2x + 5) = 14x + 35. Then subtract 4x: 14x - 4x + 35 = 10x + 35. A common mistake is forgetting to multiply 5 by 7.

AF14-004 — Correct: A — Explanation:
The service charges $28 for each stop, so that part is 28s. The $15 scheduling fee is added one time. The total charge is 28s + 15. A common mistake is treating the scheduling fee as the per-stop charge.

AF14-005 — Correct: B — Explanation:
Substitute 6 for x: f(6) = 11(6) - 20 = 66 - 20 = 46. A common mistake is adding 20 instead of subtracting it.

AF14-006 — Correct: C — Explanation:
Test the choices in y = 9x - 2. For (3, 25), 9(3) - 2 = 27 - 2 = 25, which matches the y-value. A common mistake is choosing a point that looks close without checking the equation exactly.

AF14-007 — Correct: C — Explanation:
Solve $15(x - 3) = 90$. Divide both sides by 15: $x - 3 = 6$. Add 3: $x = 9$. A common mistake is adding 3 before undoing the multiplication by 15.

AF14-008 — Correct: C — Explanation:
The pattern increases by 13 each time: 12, 25, 38, 51. Add 13 to 51: 64. A common mistake is assuming the increase is 12 because the first term is 12.

AF14-009 — Correct: C — Explanation:
$x/24 = 3$ means x divided by 24 equals 3. Multiply both sides by 24: $x = 72$. A common mistake is adding 24 and 3 instead of multiplying.

AF14-010 — Correct: B — Explanation:
Slope-intercept form is $y = mx + b$. The slope is -11 and the y-intercept is 8, so the equation is $y = -11x + 8$. A common mistake is switching the slope and y-intercept.

AF14-011 — Correct: C — Explanation:
Find the unit rate: 138 pieces ÷ 6 containers = 23 pieces per container. For 4 containers: $23 \times 4 = 92$. A common mistake is multiplying 138 by 4 without finding the rate first.

AF14-012 — Correct: B — Explanation:
Solve $100 - 15x = 40$. Subtract 100 from both sides: $-15x = -60$. Divide by -15: $x = 4$. A common mistake is losing the negative sign after subtracting 100.

AF14-013 — Correct: A — Explanation:
Combine like terms: $50y - 21y = 29y$. The constant -12 stays the same. The expression becomes $29y - 12$. A common mistake is combining the constant with the variable terms.

AF14-014 — Correct: C — Explanation:
The supply room starts with 420 forms. Workers use 14 forms each day, so after d days, 14d forms are used. The number left is $F = 420 - 14d$. A common mistake is adding 14d even though forms are being used.

AF14-015 — Correct: A — Explanation:
Substitute $x = 8$ into $y = -13x + 120$: $y = -13(8) + 120 = -104 + 120 = 16$. A common mistake is treating $-13(8)$ as positive 104.

AF14-016 — Correct: C — Explanation:
Solve $22x - 25 = 13x + 56$. Subtract 13x: $9x - 25 = 56$. Add 25: $9x = 81$. Divide by 9: $x = 9$. A common mistake is adding 25 to only one side.

AF14-017 — Correct: C — Explanation:
Solve $12x + 3 > 99$. Subtract 3: $12x > 96$. Divide by 12: $x > 8$. The only listed value greater than 8 is 9. A common mistake is choosing 8, but $12(8) + 3 = 99$, not greater than 99.

AF14-018 — Correct: B — Explanation:
The y-values increase by 14 each time x increases by 1, so the slope is 14. When $x = 0$, $y = 5$, so the y-intercept is 5. The rule is $y = 14x + 5$. A common mistake is using 5 as the slope because it is the first y-value.

AF14-019 — Correct: C — Explanation:
Distribute: $11x + 55 = 8x + 82$. Subtract $8x$: $3x + 55 = 82$. Subtract 55: $3x = 27$. Divide by 3: $x = 9$. A common mistake is forgetting to multiply 5 by 11.

AF14-020 — Correct: C — Explanation:
Substitute $a = -9$ and $b = 7$: $3a + 4b = 3(-9) + 4(7) = -27 + 28 = 1$. A common mistake is treating 3(-9) as positive 27.

AF14-021 — Correct: B — Explanation:
A positive x-coordinate and a negative y-coordinate means the point is in Quadrant IV. The point (10, -4) fits this description. A common mistake is choosing (-10, 4), which has the signs reversed.

AF14-022 — Correct: C — Explanation:
144 checks in 12 minutes means $144 \div 12 = 12$ checks per minute. In 9 minutes: $12 \times 9 = 108$. A common mistake is multiplying 144 by 9 without finding the rate first.

AF14-023 — Correct: B — Explanation:
Start with $M = 18r + 36$. Subtract 36 from both sides: $M - 36 = 18r$. Divide by 18: $r = (M - 36)/18$. A common mistake is dividing M by 18 first and then subtracting 36.

AF14-024 — Correct: C — Explanation:
Slope equals change in y divided by change in x. From (2, 5) to (8, 53), y increases by 48 and x increases by 6. The slope is $48 \div 6 = 8$. A common mistake is using 48 as the slope without dividing by the change in x.

AF14-025 — Correct: A — Explanation:
Combine like terms: $36x - 25x = 11x$, and $8 - 17 = -9$. The simplified expression is $11x - 9$. A common mistake is writing $11x + 9$ because the negative sign before 17 is missed.

AF14-026 — Correct: B — Explanation:
Substitute 12 for x: $g(12) = 12^2 - 8(12) = 144 - 96 = 48$. A common mistake is calculating 12^2 as 24 instead of 144.

AF14-027 — Correct: C — Explanation:
Start with 14. Subtract 6: 8. Then multiply by 4: 32. A common mistake is multiplying first and then subtracting 6.

AF14-028 — Correct: C — Explanation:
Solve $14(x - 5) + 8 = 92$. Distribute: $14x - 70 + 8 = 92$. Combine: $14x - 62 = 92$. Add 62: $14x = 154$. Divide by 14: $x = 11$. A common mistake is forgetting to combine -70 and +8.

AF14-029 — Correct: B — Explanation:
The line rises 6 units for every 1 unit to the right, so the slope is 6. It crosses the y-axis at -13, so the equation is $y = 6x - 13$. A common mistake is switching the slope and intercept.

AF14-030 — Correct: C — Explanation:
Use the proportion $16/24 = x/72$. Since $24 \times 3 = 72$, multiply 16 by 3: $x = 48$. A common mistake is adding 48 to the numerator because 24 becomes 72.

AF14-031 — Correct: D — Explanation:
Solve $-16x + 32 = -96$. Subtract 32 from both sides: $-16x = -128$. Divide by -16: $x = 8$. A common mistake is giving -8 because both sides contain negative numbers.

AF14-032 — Correct: B — Explanation:
The stockroom starts with 1,800 fasteners. Each job uses 90 fasteners, so j jobs use 90j fasteners. The number left is $1{,}800 - 90j$. A common mistake is adding 90j even though fasteners are being used.

AF14-033 — Correct: B — Explanation:
Test $x = 19$. In choice B, $4(19) - 15 = 76 - 15 = 61$, so the equation is true. A common mistake is choosing a nearby equation without checking both sides exactly.